US MONETARY POLICY, MARKET-IMPLIED SHORT-RATE CONDITIONS, AND THE 1-MONTH TREASURY YIELD

A Nelson-Siegel-Svensson and Stability-Aware Ridge Forecasting Study

Project objective

1. **Market sentiment analysis:** measure the gap between the Treasury curve's implied short-rate conditions and the Federal funds target midpoint.

2. 1**-month yield forecast:** gather a series of factors to forecast the 1-month yield using an optimized ridge model.

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Data: Federal Reserve Economic Data (FRED)
Treasury and target-rate series: Board of Governors of the Federal Reserve System

Executive summary

This project links the US Treasury yield curve with Federal Reserve policy and then uses the derived information in a next-month 1-month Treasury-yield forecasting model. Monthly Nelson-Siegel-Svensson (NSS) curves produce an implied instantaneous short rate. The difference between that rate and the Federal funds target midpoint defines the policy spread ($policy\ spread = \text{NSS short rate} - midpoint\ of\ policy\ target\ rate\ ranges$).

After the first part, the second part of the study utilizes the Ridge model to factor in momentum, short-rate change, and policy-spread information for the forecast of the 1-month treasury yield. After that, the model is compared with the no-change benchmark (please refer to the below).

Key components:

- $Policy\ spread = \text{NSS short rate} - policy\ midpoint$
- RMSE refers to root mean square error, which is used to evaluate the forecasting power of a model.
- MAE is the mean absolute error, which is also a measure of the model's forecasting power.
- No-change benchmark is $\hat{y}_{t+1}^{benchmark} = y_t$, meaning that the current yield estimates the next-month yield, where y_t is the 1-month yield, t is the current month, t + 1 is the next month, and $\hat{y}_{t+1}^{benchmark}$ is the expected next-month yield.

Summary of the out-of-sample backtest and comparison:

Metric	Model	Benchmark	Improvement
RMSE	0.201	0.207	2.589%
MAE	0.111	0.112	0.844%

Main conclusion

- The model outperforms the no-change benchmark modestly in terms of RMSE and MAE.
- This means my model has better forecasting power than the no-change benchmark because the forecast error of my model is lower.
- Policy spread reflect the market sentiment of short rates relative to the policy midpoint, and the spread has forecasting power of the 1-month yield.

1. Research design

1.1 Part I: policy spread

Part I compares the short rate implied by the fitted Treasury curve with the Federal funds target midpoint. The objective is to assess whether market-implied short-rate conditions are above, below, or broadly aligned with the contemporaneous policy setting.

1.2 Part II: 1-month Treasury-yield forecast

Part II uses yield momentum and policy-related variables to forecast the next-month change in the observed 1-month Treasury yield. Candidate specifications are evaluated against a no-change benchmark under rolling, multi-period validation.

2. Data

Daily Treasury constant-maturity yields and Federal funds target-range data are retrieved through the FRED API and converted to month-end observations using the last available value in each month. The target midpoint is the average of the upper and lower target-range limits.

$$Policy\ midpoint_t = \frac{(Upper\ limit_t + Lower\ limit_t)}{2}$$

Data block	Content
Treasury curve	1M, 3M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 20Y, and 30Y constant-maturity yields
Policy	Federal funds target upper limit, lower limit, and calculated midpoint
Frequency	Daily observations converted to month-end
Study period	January 2009 to October 2026

3. NSS curve and policy spread

3.1 NSS specification

$$y(\tau) = \beta_0 + \beta_1 L_1(\tau) + \beta_2 L_2(\tau) + \beta_3 L_3(\tau)$$

The NSS model summarizes the curve using **a long-run level,** a **slope component, and two curvature components**.

3.2 Implied short rate

Taking the maturity limit as τ approaches zero gives:

$$\text{NSS short rate} = r(0) = \lim_{\tau \to 0} y(\tau) = \beta_0 + \beta_1$$

This is a theoretical instantaneous rate derived from the fitted curve. NSS short rate can be a good reflection of what the market thinks the short rate should be over the infinitely small maturity τ.

3.3 Policy spread

$$Policy\ spread_t = NSS\ short\ rate_t - Policy\ midpoint_t$$

Spread	Interpretation
Positive	The market expects more policy tightening than the actual target rate midpoint does.
Near zero	Market-implied conditions are broadly aligned with current policy.
Negative	The market expects more policy easing than the actual target rate midpoint does.

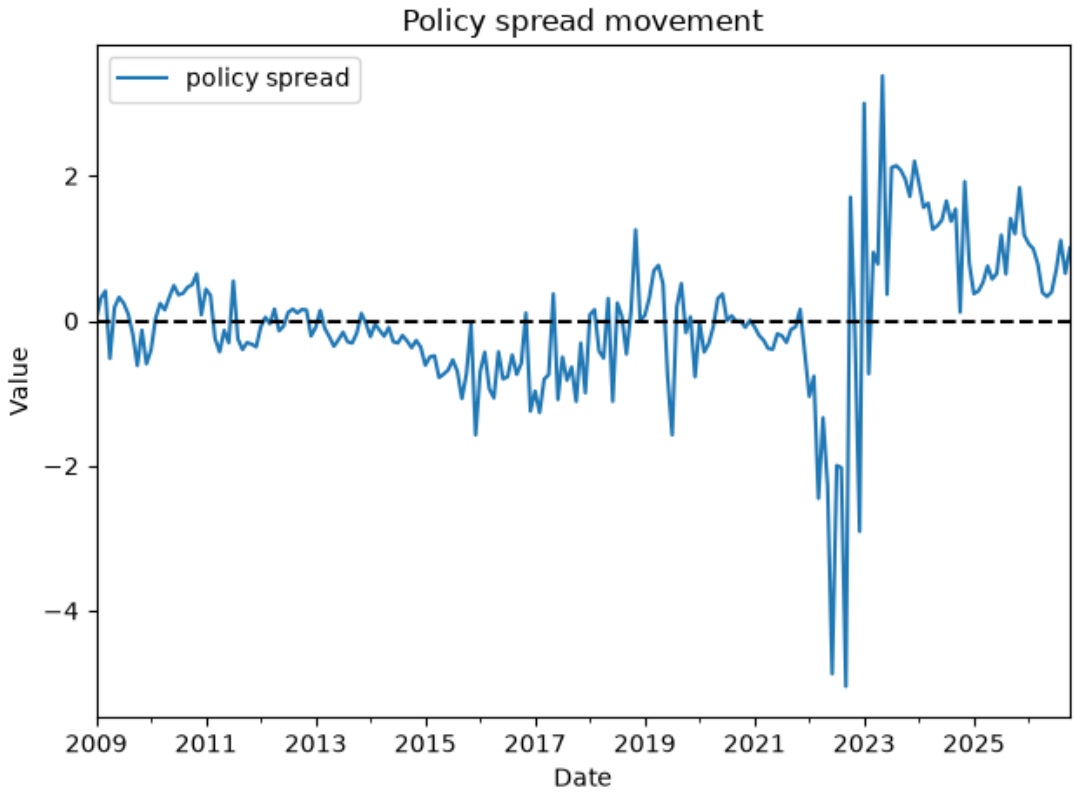


Figure 1. Policy-spread movement over time
Source: Calculations using FRED data.

The policy spread became more volatile from 2021 onward than before. From 2023 to 2026, the spread remains positive, suggesting that the **market expected more policy tightening than the policy midpoint did**.

Potential reasons for the higher-than-usual policy spread might include:

- Accelerating US GDP than expected.
- Higher CPI than expected.
- Unresolved US-Iran conflict.
- AI integration that increases productivity and inflation expectation.

Another interesting fact can be observed from this figure. Mean reversion is spotted. The curve moves back to the horizontal line y = 0 historically. Now, the recent spread is becoming smaller, seeming that it is returning to the horizontal line. This might reflect that the market expectation will gradually align with the policy target.

3.4 Implication of the policy spread

One may argue that the long-term yield minus the short-term yield spread can reflect the market expectation about inflation growth and term premium and that there is no need to consider the policy spread.

The key difference is that policy spread measures the market-policy alignment, not expectations of inflation growth and term premium. The policy spread compares the market expected short yield with the midpoint of the policy target rate ranges. By knowing policy spread, we can understand whether the market prices in the potential rate hike more than those implied by the current policy target midpoint. That is why finding the fitted short value still has its economic meaning.

4. Forecasting methodology

Here starts the second part of my study: fit the Ridge model using a series of factors, including the policy spread calculated in part 1, to predict $\boldsymbol{\Delta y_{t+1}^{1M}}$.

4.1 Target variable

$$\Delta y_{t+1}^{1M} = y_{t+1}^{1M} - y_t^{1M}$$

The model predicts the next-month change in the observed 1-month Treasury yield. The final yield forecast adds the predicted change to the current observed yield.

4.2 Candidate predictor groups

Below are the possible factors that are tested for the model.

Group	Economic content
Momentum	Current and lagged observed 1-month yield changes
Smoothed momentum	Current change, 3-month mean change, and recent volatility
Policy and momentum	Yield momentum, policy-midpoint change, and policy spread
Policy dynamics	Policy-midpoint change, policy spread, and policy-spread change
NSS short-rate dynamics	Yield momentum, NSS short-rate change, and policy spread

4.4 Selected model formula

The final model reported in the project uses the following predictors:

$$X_t = \begin{bmatrix} \Delta y_t^{1M} \\ \Delta m_t \\ s_t \\ \Delta s_t \end{bmatrix}$$

where:

- Δy_t^{1M}: observed 1-month yield change
- Δm_t: policy midpoint change
- s_t: policy spread
- Δs_t: policy-spread change

The raw forecast is

$$\widehat{\Delta y}_{t+1}^{raw} = \hat{\beta}_0 + \hat{\beta}_1 Z(\Delta y_t^{1M}) + \hat{\beta}_2 Z(\Delta m_t) + \hat{\beta}_3 Z(s_t) + \hat{\beta}_4 Z(\Delta s_t)$$

4.3 Rolling Ridge estimation

For each forecast date, predictors are standardized using historical training data only. Ridge coefficients minimize squared forecast errors plus an L2 penalty:

$$\min_{\beta}\left[\sum_{i=t-48}^{t-1}\left(\Delta y_{i+1}^{1M}-\left(\beta_0+\sum_{j=1}^{4}\beta_j\,Z\left(X_{j,i}\right)\right)\right)^2+180\sum_{j=1}^{4}\beta_j^2\right]$$

Where:

- 180 is the selected alpha to penalize the excessively large coefficients.
- β_j is the beta of the factors.
- $Z\left(X_{j,i}\right)$ is the standardized predictor value of the factors.

Then the minimized factor coefficients are:

$$\hat{\gamma}=(Z^{\top}Z+180I_4)^{-1}Z^{\top}(y-\bar{y}1)$$

where:

$$\hat{\gamma}=\begin{bmatrix}\hat{\gamma}_1\\ \hat{\gamma}_2\\ \hat{\gamma}_3\\ \hat{\gamma}_4\end{bmatrix}$$

4.4 Forecast weight

$$\widehat{\Delta y}_{t+1}^{weighted}=0.25\times\widehat{\Delta y}_{t+1}^{raw}$$

Where:

- 0.25 is the selected weight to optimize the model.

4.4 Signal threshold

The weighted forecast is used only if its absolute value is at least **1.675 basis points**:

$$\widehat{\Delta y}_{t+1}^{used}=\begin{cases}\widehat{\Delta y}_{t+1}^{weighted}, & \left|\widehat{\Delta y}_{t+1}^{weighted}\right|\geq 0.01675\\ 0, & \left|\widehat{\Delta y}_{t+1}^{weighted}\right|<0.01675\end{cases}$$

Hence, $\widehat{\Delta y}_{t+1}^{used}$ is what my model calculates in the end.

The expected next-month 1-month yield is given as below:

$$\hat{y}_{t+1}^{1M}=y_t^{1M}+\widehat{\Delta y}_{t+1}^{used}$$

Where:

- $\hat{y}_{t+1}^{1M}$ is the model-implied next-month 1-month yield
- $\widehat{\Delta y}_{t+1}^{used}$ is the model-implied change of next-month 1-month yield
- y_t^{1M} is the current 1-month yield

5. Results

5.1 Validation / trained-model output retained from the file

Item	Value
Factors	Observed 1M yield change; 3M mean change; short-rate change; policy spread
Window	Expanding window
Alpha	180.0
Weight	1.0
Threshold	4.375 bps
Validation model RMSE	0.3391
Validation benchmark RMSE	0.4819
Validation model MAE	0.2725
Validation benchmark MAE	0.3718

These values describe the selected or trained validation specification saved in the source document. They should not be mixed with the final out-of-sample metrics below.

5.2 Out-of-sample backtest

Metric	Model	No-change benchmark	Improvement
RMSE	0.201	0.207	2.589%
MAE	0.111	0.112	0.844%

The model has lower errors than the no-change benchmark. Therefore, it has more predicting power. Meanwhile, it is worth noting that the improvement is modest.

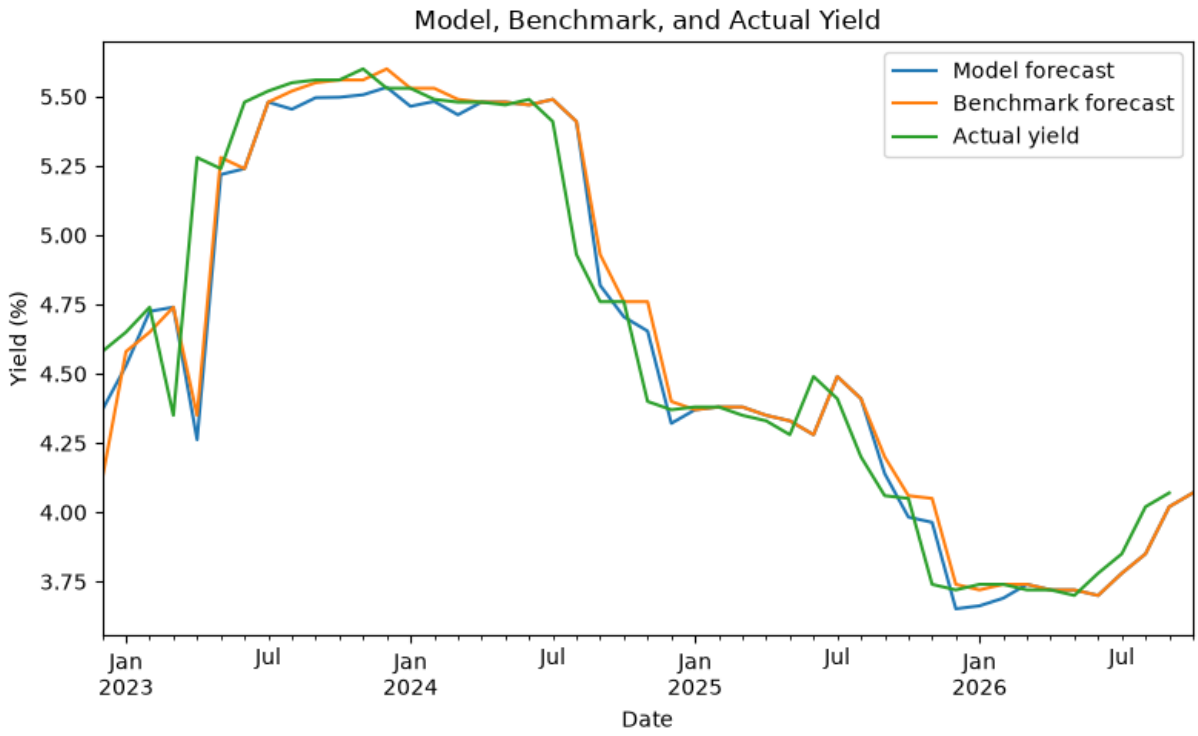
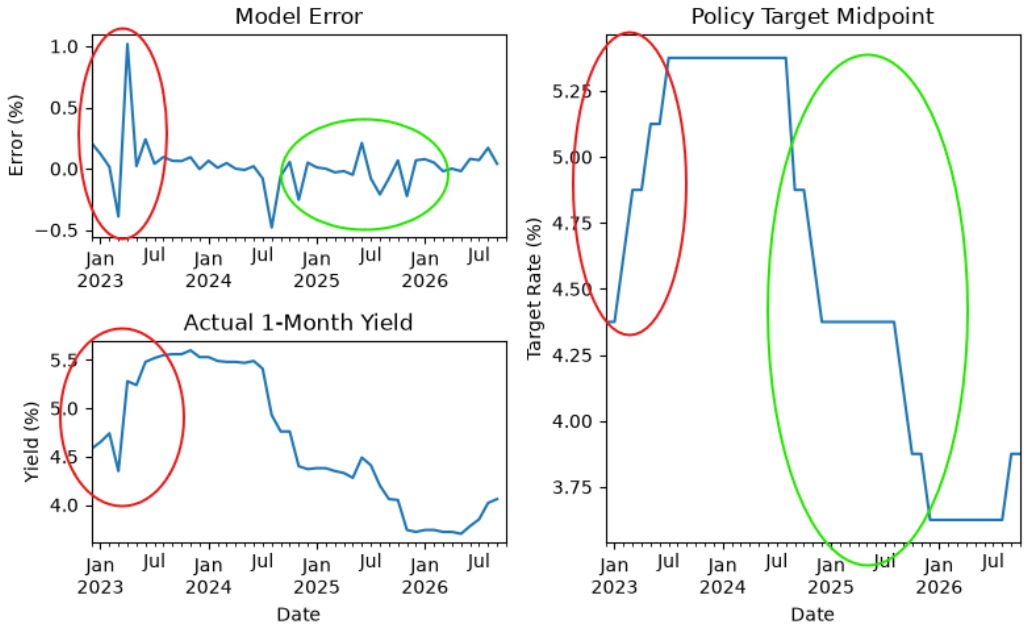


Figure 2. Actual 1-month yield, Ridge forecast, and no-change benchmark
Source: Author calculations using FRED data.

The figure visualizes how closely the model and the benchmark follow the actual 1-month yield.

5.3 Model error analysis



The model's largest forecast error occurred in early 2023, when the observed 1-month Treasury yield changed abruptly during a period of rapid policy tightening and unusual short-term market conditions. The 1% height of the model error could be explained by the rapid hike of the target rate.

In contrast, errors remained relatively small during the easing period from late 2024 to early 2026 because the policy midpoint and 1-month yield **declined more smoothly**. There could be two explanations for the phenomenon:

1. The model is better at predicting policy easing than tightening; or
2. The model captures graduate policy-driven trends better than sudden market reversal.

8. Conclusion

The project develops a coherent two-stage framework. NSS curve fitting provides an economically interpretable short-rate measure and policy spread. A rolling Ridge model then tests whether that information improves next-month forecasts of the observed 1-month Treasury yield. The final backtest beats the no-change benchmark under both RMSE and MAE, but only modestly. It is concluded that policy and NSS indicators provide incremental forecasting information while the current yield remains a very strong baseline.